

No. of Functional Features Included in the Solution

Date	25 May 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI Loan Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Allows authorized users to log into the system securely.	Login Module	Done	High
2	Applicant Data Entry	Collects applicant details through an input form.	Frontend (HTML/Flask)	Done	High
3	Data Preprocessing	Cleans, encodes, and scales the applicant data before prediction.	Data Processing	Done	High
4	Loan Prediction	Predicts loan approval using the trained XGBoost model.	ML Model	Done	High
5	Result Display	Displays whether the loan is Approved or Rejected.	Prediction Module	Done	High
6	Model Evaluation	Compares multiple ML models and selects the best-performing model.	Model Evaluation	Done	Medium
7	Report Generation	Displays prediction results and model performance. Integrates the machine learning model with a web-based interface.	Reporting Module	Done	Medium
8	Flask Web Application	Integrates the machine learning model with a web-based interface.	Backend	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Login Page, Applicant Form
2	Backend / Logic	3	Data Preprocessing, Prediction, Result Processing
3	Database / Storage	1	Loan Prediction Dataset (CSV)
4	API / Integration	1	Flask Integration with ML Model
5	Security / Authentication	1	User Login Authentication